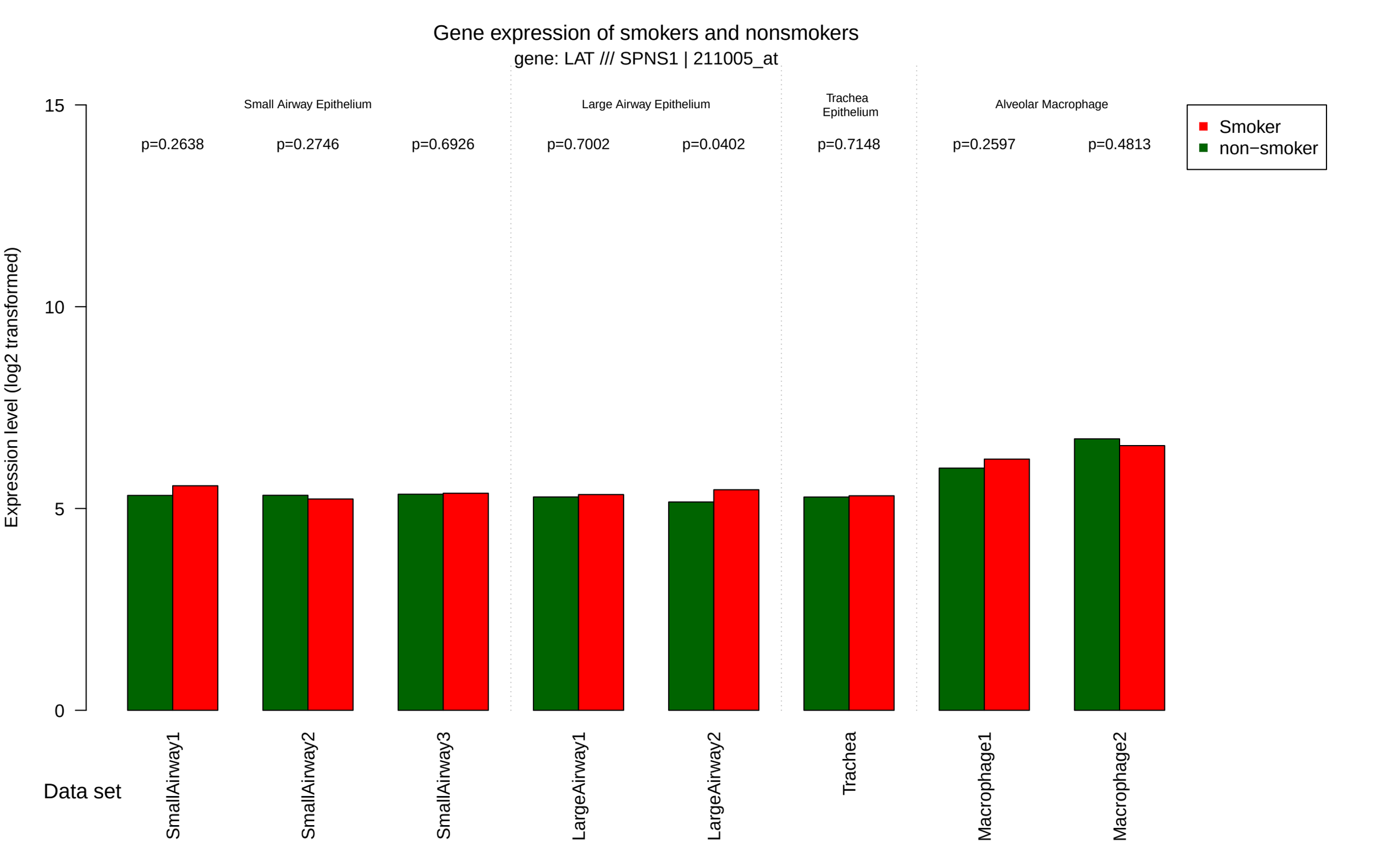
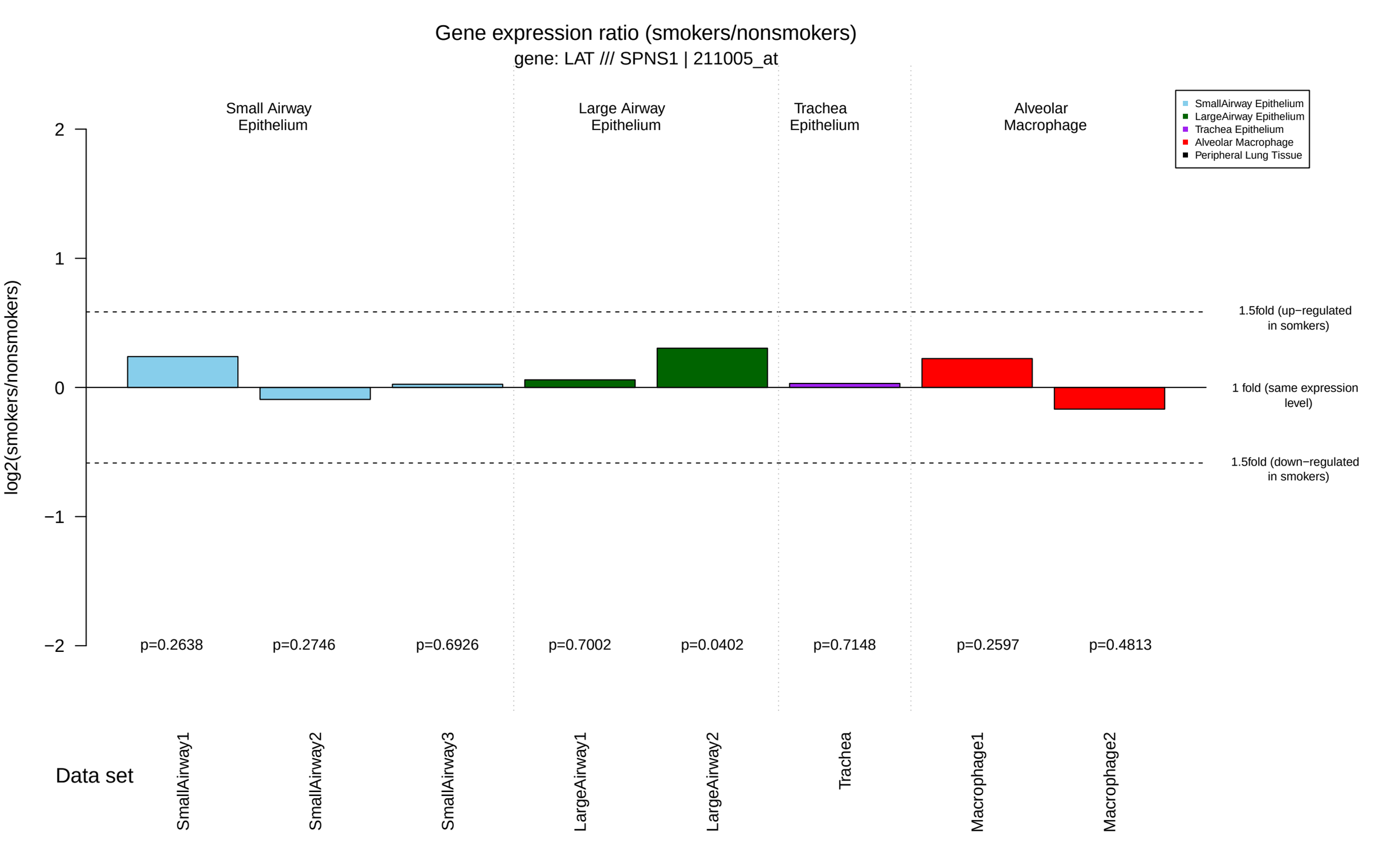
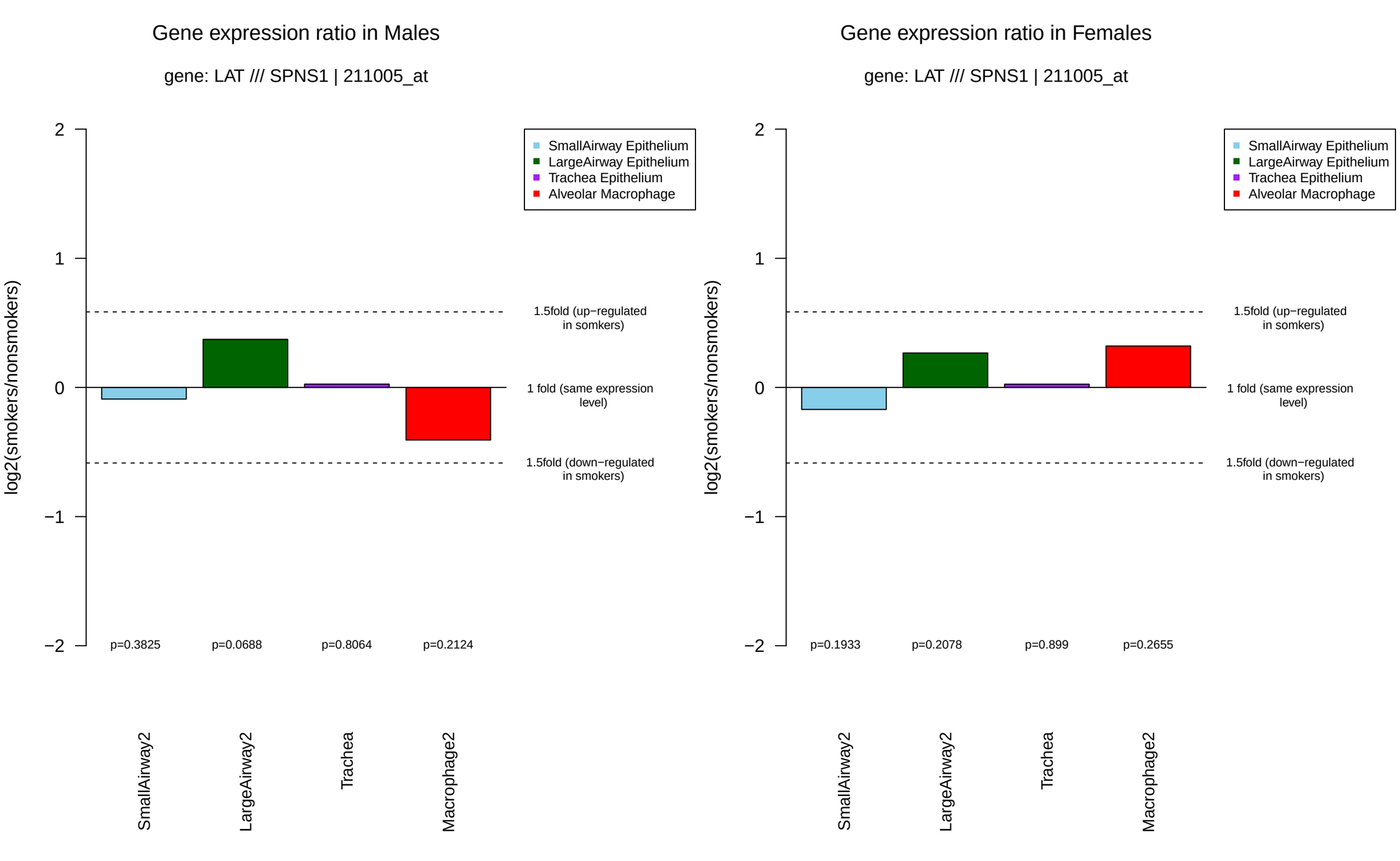
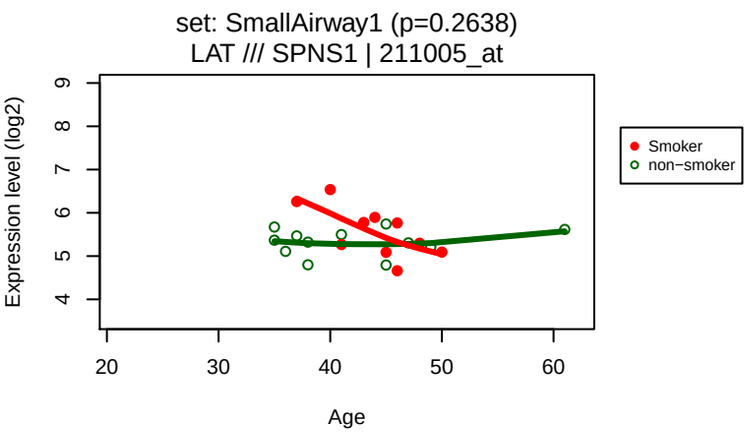
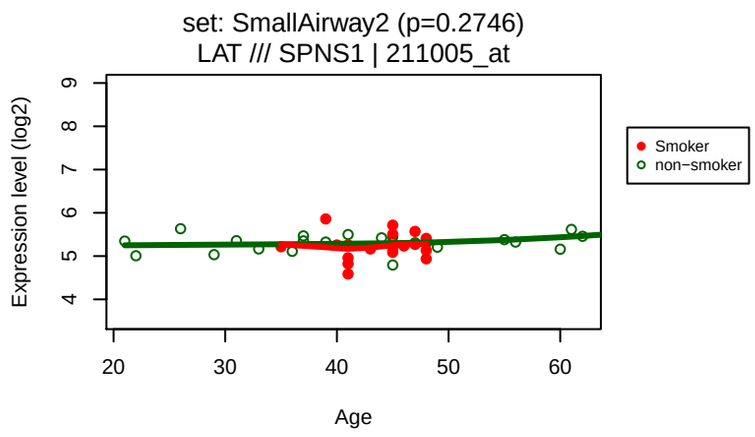
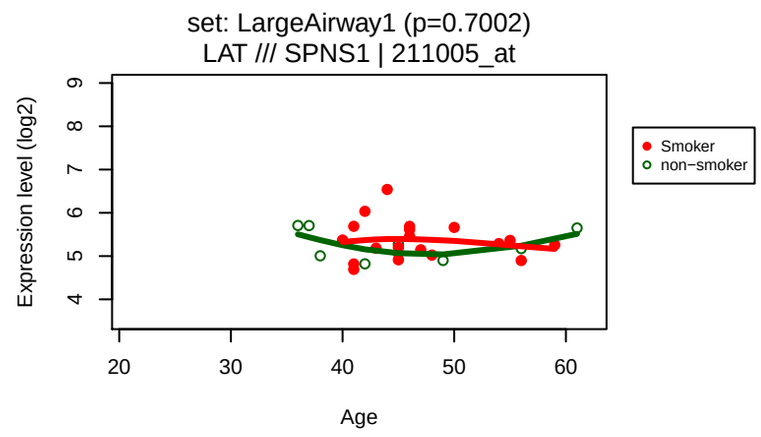

Gene expression of smokers and nonsmokers
gene: LAT /// SPNS1 | 211005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2638
p=0.2746
p=0.6926
p=0.7002
p=0.0402
p=0.7148
p=0.2597
p=0.4813
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

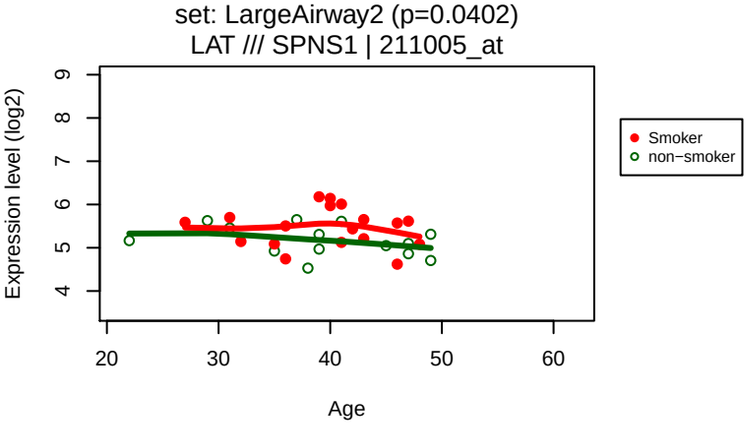

Gene expression ratio (smokers/nonsmokers)
gene: LAT /// SPNS1 | 211005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2638
p=0.2746
p=0.6926
p=0.7002
p=0.0402
p=0.7148
p=0.2597
p=0.4813
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

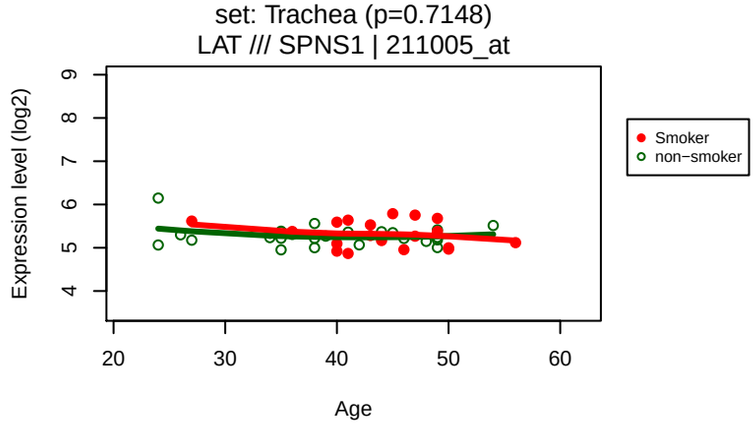

Gene expression ratio in Males
gene: LAT /// SPNS1 | 211005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3825
p=0.0688
p=0.8064
p=0.2124
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAT /// SPNS1 | 211005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1933
p=0.2078
p=0.899
p=0.2655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

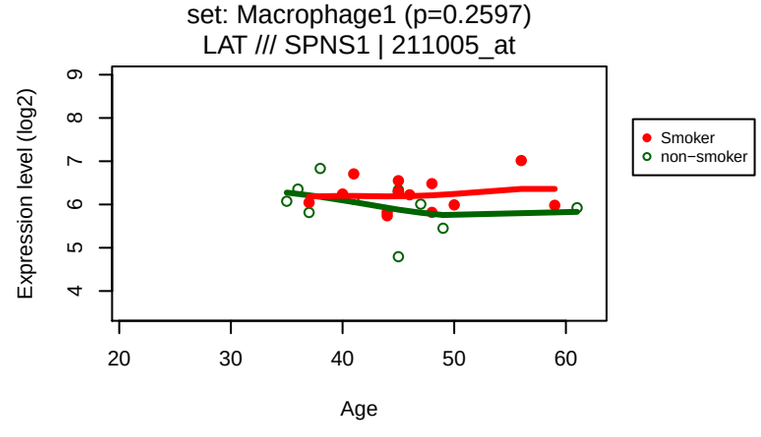

set: SmallAirway1 (p=0.2638)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker

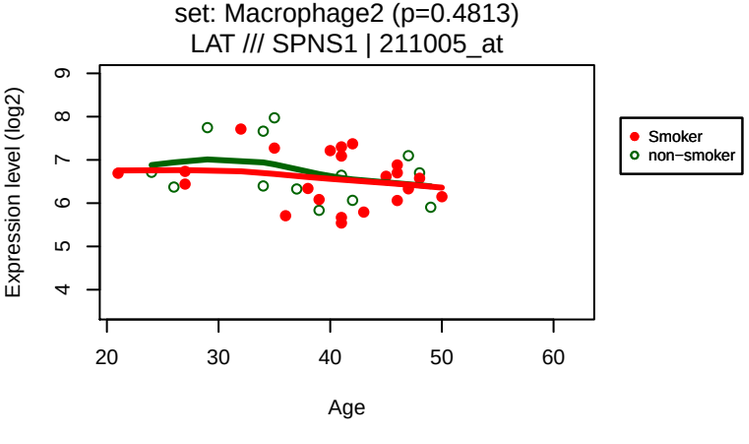

set: SmallAirway2 (p=0.2746)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker

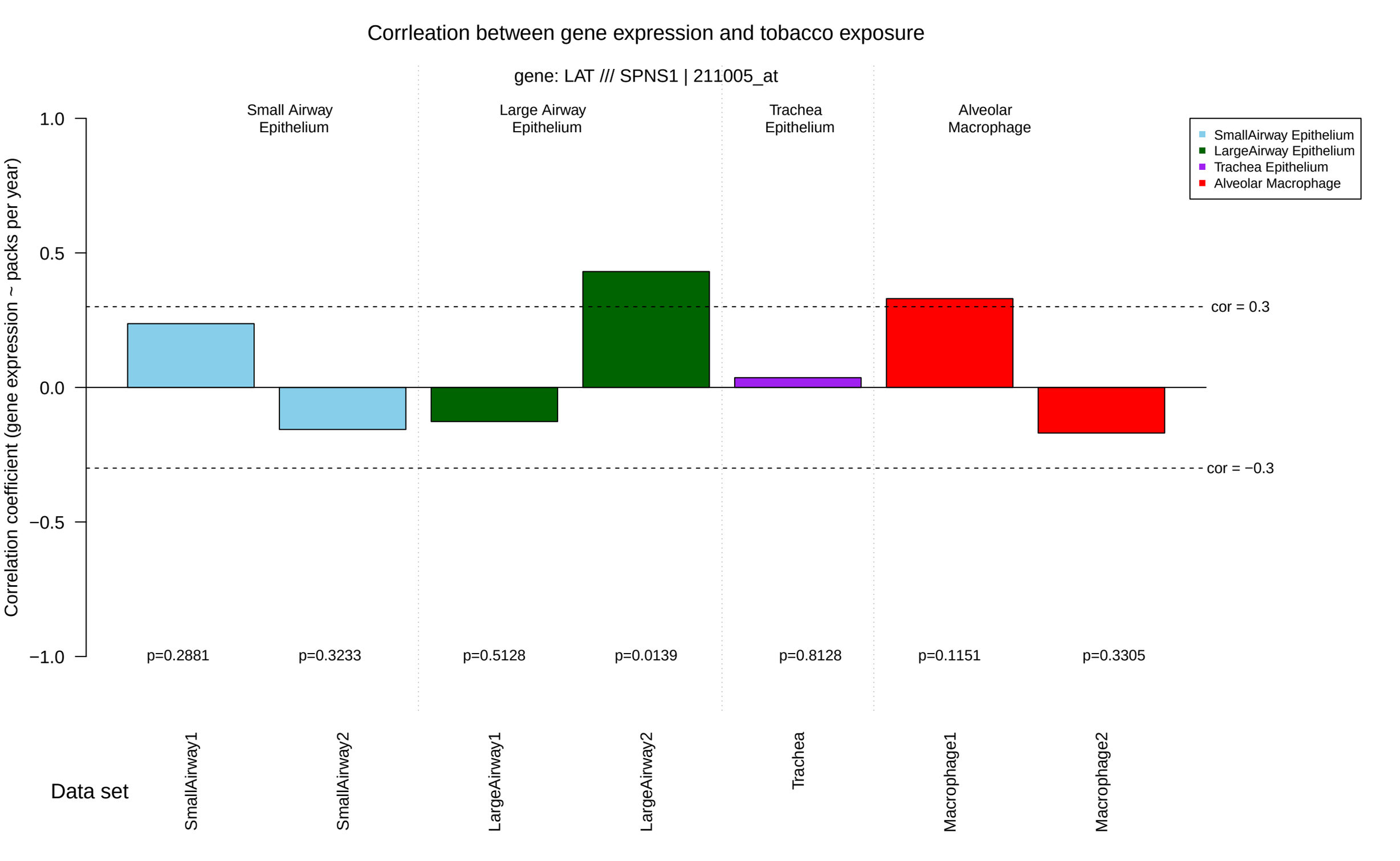

set: LargeAirway1 (p=0.7002)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0402)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7148)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2597)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4813)
LAT /// SPNS1 | 211005_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAT /// SPNS1 | 211005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2881
p=0.3233
p=0.5128
p=0.0139
p=0.8128
p=0.1151
p=0.3305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2